Case 1

Done	
Status	Cases

1. Two memory systems

Long-term memory comes in two forms — **explicit** (conscious recall) and **implicit** (unconscious, expressed in performance) — that depend on different brain systems. In every system, storage is a lasting change in the strength of synaptic connections.

- **Explicit (declarative):** facts (semantic) and events (episodic) about people, places, objects
  - Needs medial temporal lobe + hippocampus; ultimately stored in neocortex
- **Implicit (nondeclarative):** recalled in action, not in words
  - Priming → neocortex
  - Skills and habits (procedural) → striatum
  - Associative learning: emotional responses → amygdala; skeletal musculature → cerebellum
  - Nonassociative learning (habituation, sensitization) → the reflex pathways themselves
- The systems work together; repetition turns explicit learning into implicit (learning to drive → automatic)

Evidence: H.M. — after medial temporal lobe removal, no new conscious memories, yet normal learning of new motor skills → the two systems are separable.

2. Elementary plasticity: the Aplysia gill-withdrawal reflex

The three simplest forms of implicit learning are bidirectional changes in transmitter release at the **same** sensory → motor synapses: habituation weakens them, sensitization and classical conditioning strengthen them.

The reflex circuit

1. Siphon touch → excites siphon mechanoreceptor sensory neurons → they fire action potentials
2. Sensory neurons → release glutamate onto interneurons and gill motor neurons → fast EPSPs in both
3. Interneurons → also excite the motor neurons → EPSPs from sensory neurons and interneurons sum in the motor neurons
4. Motor neurons → fire strongly → gill muscles contract → gill withdraws

- Learning changes several of these synapses at once → memory is distributed across the circuit

2.1 Habituation — homosynaptic depression

Learning to ignore a repeated harmless stimulus: the stimulated pathway itself releases less transmitter.

1. Siphon touched repeatedly → sensory neuron fires the same action potential each time
2. Each action potential → fewer synaptic vesicles released from the sensory terminals (same at interneuron → motor synapses)
3. Less glutamate → smaller EPSPs in interneurons and motor neurons
4. Smaller summed excitation → motor neurons fire less → weaker gill withdrawal

- **Homosynaptic:** caused by activity in the pathway itself; no modulatory cell needed
- Recovers within about an hour; repeated sessions → long-term habituation = loss of synapses (§4.3)

Evidence: quantal analysis — less glutamate released, postsynaptic receptor sensitivity unchanged → the change is presynaptic.

2.2 Sensitization — heterosynaptic presynaptic facilitation

A noxious stimulus enhances responses to other, harmless stimuli: modulatory interneurons release serotonin onto the sensory terminals and boost their transmitter release.

Getting serotonin to the synapse

1. Tail shock → activates tail sensory neurons → they excite facilitating (modulatory) interneurons
2. Facilitating interneurons → release serotonin (5-HT) at axo-axonic synapses on the siphon sensory terminals

Pathway 1 — cAMP–PKA broadens the action potential

1. 5-HT binds a metabotropic receptor on the sensory terminal → activates the G protein Gs
2. Gs → stimulates adenylyl cyclase → cyclase converts ATP to cAMP
3. cAMP binds the regulatory subunits of PKA → releases the catalytic subunits → PKA active
4. PKA phosphorylates K+ channels → channels close → less outward K+ current
5. Less repolarizing current → action potential lasts longer
6. Longer depolarization → voltage-gated Ca2+ channels stay open longer → more Ca2+ influx
7. More Ca2+ → more vesicles fuse → more glutamate → larger motor-neuron EPSP → stronger withdrawal

Pathway 2 — PKC makes more vesicles ready

1. 5-HT binds a second receptor type → activates the G protein Gq
2. Gq → stimulates phospholipase C → produces diacylglycerol (DAG)
3. DAG → activates PKC
4. PKC phosphorylates presynaptic proteins → vesicles move from the reserve pool to the releasable pool at the active zone → more release per action potential

- **Heterosynaptic:** driven by a different (modulatory) pathway
- Same synapses as habituation → modifiable synapses are bidirectional, via different mechanisms (homo- vs heterosynaptic)
- One shock → minutes; repeated shocks → days to weeks (§4)
- Dishabituation = sensitization overriding habituation

2.3 Classical conditioning — activity-dependent presynaptic facilitation

When a weak CS (siphon touch) just precedes a strong US (tail shock), facilitation is larger than sensitization and confined to the CS pathway. Adenylyl cyclase is the **coincidence detector**.

1. CS: siphon touch → siphon sensory neuron fires an action potential
2. Action potential → Ca2+ enters the sensory terminal through voltage-gated channels
3. Ca2+ binds calmodulin → Ca2+/calmodulin binds adenylyl cyclase → cyclase is "primed"
4. US shortly after: tail shock → facilitating interneurons release 5-HT onto the same terminals
5. 5-HT → Gs → the primed cyclase responds more vigorously → more cAMP than 5-HT alone produces
6. More cAMP → more PKA activity → greater presynaptic facilitation (Pathway 1, §2.2)
7. Only terminals that fired just before the US were primed → enhancement confined to the CS pathway → CS alone now evokes strong withdrawal

- Order matters: 5-HT before Ca2+ → cyclase not primed → no conditioning → the CS must precede and predict the US
- = sensitization mechanism + a coincidence detector reading CS (Ca2+) and US (5-HT)
- Plus a postsynaptic component: Ca2+ entry into the strongly excited motor neuron (LTP-like, §3.2)

3. Associative plasticity in mammals: hippocampal LTP and LTD

Long-term potentiation (LTP) is a long-lasting increase in synaptic strength induced by brief high-frequency stimulation (a tetanus). It is a family of mechanisms, all homosynaptic (induced by activity in the pathway that is potentiated), that differ in receptors and signaling; at the Schaffer collateral synapse the NMDA receptor makes it associative.

The hippocampal circuit (perforant pathways from entorhinal cortex)

1. Entorhinal cortex layer II neurons → perforant path → excite dentate granule cells
2. Granule cells → mossy fibers → excite CA3 pyramidal cells
3. CA3 cells → Schaffer collaterals → excite the proximal dendrites of CA1 pyramidal cells
4. In parallel, entorhinal layer III neurons → direct perforant path → excite the distal dendrites of CA1
5. CA1 (main output) → subiculum and back to entorhinal cortex

- Steps 1–3 = indirect **trisynaptic** pathway; step 4 = **direct** pathway → CA1 may compare processed with direct input

3.1 Mossy fiber LTP — nonassociative, presynaptic

1. Tetanus → many action potentials in mossy fiber terminals → large Ca2+ influx into the presynaptic terminal
2. Ca2+/calmodulin → activates adenylyl cyclase → cAMP rises
3. cAMP → activates PKA in the terminal
4. PKA phosphorylates RIM1α, a vesicle protein that organizes exocytosis → more glutamate released per action potential → larger EPSP in CA3

- No postsynaptic activity needed → **nonassociative**; NMDA receptors play no role
- Same logic as Aplysia sensitization (§2.2): presynaptic cAMP–PKA, boosted by a modulator (norepinephrine → β-adrenergic receptors → adenylyl cyclase, in

place of 5-HT)

Evidence: NMDA blocker (APV) leaves it intact; PKA inhibitor blocks it → presynaptic cAMP–PKA, not NMDA.

3.2 Schaffer collateral LTP — associative, NMDA-dependent

The NMDA receptor opens only when glutamate is bound **and** the postsynaptic membrane is depolarized, so it detects coincident pre- and postsynaptic activity and admits the Ca2+ that triggers LTP.

Normal low-frequency transmission

1. Action potential in the CA3 terminal → glutamate released
2. Glutamate binds both AMPA and NMDA receptors on the CA1 dendritic spine
3. AMPA receptors open → Na+ in, K+ out → EPSP
4. NMDA receptors bind glutamate but their pore is blocked by extracellular Mg2+ at negative potentials → they pass little current

Induction (postsynaptic)

1. Tetanus → many AMPA receptors open → large summed depolarization → postsynaptic action potential
2. Depolarization → repels Mg2+ out of the NMDA pore (works only where glutamate is bound)
3. Unblocked NMDA receptors → conduct Na+, K+ and, critically, Ca2+ → Ca2+ rises in the spine
4. Ca2+/calmodulin → activates CaMKII, PKC and the tyrosine kinase Fyn → LTP induced

Expression — postsynaptic

1. Kinases (PKC) phosphorylate the tails of AMPA receptors held in recycling endosomes → these vesicles fuse with the spine membrane → more AMPA receptors at the synapse
2. Same amount of glutamate → larger AMPA current → bigger EPSP

Expression — presynaptic

1. Postsynaptic Ca2+-activated messengers → the cell releases a retrograde messenger (perhaps NO, made by NO synthase) → it diffuses to the presynaptic terminal
2. Retrograde messenger → activates kinases only in terminals that were just active → more glutamate release

- **Silent synapses** = synapses with only NMDA receptors (silent at rest because of Mg2+ block). During LTP, depolarization from other, AMPA-containing synapses spreads to them → Mg2+ relieved → Ca2+ enters → AMPA receptors inserted (step 5) → synapse unsilenced ("AMPAfication")
- Induction is always postsynaptic; expression varies with stimulation pattern: a stronger tetanus also opens L-type Ca2+ channels and adds presynaptic expression

Evidence: pairs connected only by silent synapses give no response at rest, an NMDA response at +30 mV, and normal AMPA EPSPs after LTP → LTP adds postsynaptic AMPA receptors.

3.3 Hebbian properties

Because induction needs strong postsynaptic depolarization at a synapse with glutamate bound, Schaffer LTP follows Hebb's rule: A repeatedly helping to fire B strengthens A → B.

- **Cooperativity:** many weak inputs could fire together to depolarize enough to expel Mg2+ → significant events are stored
- **Associativity:** a weak input active together with a strong one is potentiated (the strong input's depolarization spreads to the weak synapse) → cellular analog of classical conditioning
- **Synapse specificity:** an inactive synapse has no glutamate on its NMDA receptors → no Ca2+ entry despite depolarization → not potentiated
- Two associative steps in series: postsynaptic NMDA receptor + retrograde signal acting only on active terminals → same logic as Aplysia conditioning (§2.3) and amygdala (§6.2)

Spike timing-dependent plasticity

1. Presynaptic spike → glutamate binds NMDA receptors → channel gate open but pore still Mg2+-blocked
2. Postsynaptic spike a few ms later, during the EPSP → Mg2+ expelled while glutamate is still bound → large Ca2+ influx → LTP
3. Postsynaptic spike first → Mg2+ relieved while no glutamate is bound → gate closed → little Ca2+ → no LTP (LTD instead); spikes far apart → no change

3.4 Long-term depression (LTD)

LTD weakens synapses, preventing saturation and allowing old associations to be revised. In the hippocampus it is induced by prolonged low-frequency stimulation and also needs NMDA receptors — the **size of the Ca2+ signal** decides the direction.

1. Prolonged low-frequency stimulation → only modest postsynaptic depolarization → Mg2+ block of NMDA receptors barely relieved
2. Little Ca2+ enters → postsynaptic Ca2+ rises only slightly → too little to activate CaMKII
3. Small Ca2+ rise → activates calcineurin, a Ca2+-dependent phosphatase with higher Ca2+ affinity than CaMKII
4. Calcineurin → activates further phosphatases → they dephosphorylate the GluA1 subunit of AMPA receptors (while PKC phosphorylates GluA2)
5. These changes → AMPA receptors endocytosed → fewer receptors in the membrane → smaller EPSP

- Large Ca2+ rise → kinases → LTP; small rise → phosphatases → LTD
- Anti-Hebbian rule: synapses that don't help fire the cell (or fire after it) are weakened
- Cerebellar LTD works differently (§7.6)

Evidence: mice unable to make hippocampal LTD learn a water-maze platform normally but keep searching the old spot when it moves → LTD supports flexible relearning.

4. From short- to long-term memory: a common genetic switch

Short-term memory = covalent modification (phosphorylation) of existing proteins at existing synapses. Repetition converts it into long-term memory (**consolidation**), which needs gene transcription and new proteins and is expressed as growth or loss of synapses. The switch — cAMP → PKA → CREB → chromatin → genes — is shared by Aplysia, flies and mammals, implicit and explicit memory.

4.1 The CREB switch

1. Repeated, spaced 5-HT (repeated tail shocks) → prolonged rise in cAMP in the sensory neuron
2. Sustained cAMP → sustained PKA activity → PKA catalytic subunits translocate into the nucleus
3. PKA also (indirectly) activates MAPK → MAPK also enters the nucleus
4. In the nucleus, PKA phosphorylates **CREB-1** (activator), already bound to CRE promoter elements → CREB-1 can now activate transcription
5. MAPK phosphorylates **CREB-2** (repressor) → CREB-2 leaves the promoter → repression lifted
6. Activation + de-repression → transcription of cAMP-inducible genes (e.g., ubiquitin hydrolase, C/EBP; §4.3)

- Activator + repressor at the first step → adjustable threshold for long-term storage (attention, mood, context)

4.2 Chromatin: epigenetic control

Whether a gene can be read depends on how tightly its DNA is wound around histones (nucleosomes); covalent changes to chromatin = epigenetic regulation.

Opening chromatin (acetylation)

1. Basal state: CREB-1 sits on the CRE, but the TATA box (where RNA polymerase II is recruited) is wrapped in a nucleosome → transcription blocked

2. Phosphorylated CREB-1 → recruits **CBP** (CREB-binding protein) to the promoter
3. CBP, an acetyltransferase → acetylates lysines in histone tails → neutralizes their positive charge
4. Histones bind negatively charged DNA less tightly; activators that recognize acetylated histones reposition the nucleosome → TATA box exposed
5. CBP also recruits RNA polymerase II → transcription starts

Closing chromatin (DNA methylation)

1. DNA methyltransferases → methylate cytosines at CpG sites
2. Methyl-CpG-binding proteins bind → recruit histone deacetylases (HDACs)
3. Histones deacetylated (with CREB-2) → chromatin closes → gene repressed

- Less acetylation (defective CBP) → impaired late LTP and long-term memory; HDAC inhibitors → enhanced memory

4.3 What the new genes do

A persistently active kinase

1. CREB-1 → induces ubiquitin carboxyterminal hydrolase
2. Hydrolase → boosts ubiquitin–proteasome degradation of PKA's regulatory subunits (PKA = two regulatory subunits inhibiting two catalytic ones)
3. Freed catalytic subunits → active without cAMP → keep phosphorylating release proteins and CREB-1 for about a day after cAMP returns to baseline

- = the simplest long-term mechanism: a second-messenger kinase made constitutively active

Growth of new synapses

1. CREB-1 → induces the transcription factor C/EBP
2. C/EBP → dimerizes (with itself or activating factor) → turns on downstream genes
3. Their products → new presynaptic terminals in the sensory neuron + dendritic growth in the motor neuron → more synapses

- Long-term sensitization → more terminals; long-term habituation → fewer (pruning)
- Short-term = functional change in existing synapses; long-term = change in their number

4.4 Early vs late LTP

The hippocampus uses the same two-stage logic.

- **Early LTP:** one train; uses only modifications of existing proteins (§3.2); no protein synthesis, no PKA

Late LTP

1. Repeated tetani → repeated Ca2+ influx through NMDA receptors
2. Ca2+/calmodulin → recruits adenylyl cyclase → cAMP → PKA
3. PKA → activates MAPK → both reach the nucleus → CREB-1 phosphorylated
4. CREB-1 → transcription of CRE target genes → new proteins
5. New proteins → new presynaptic release sites and spines, apposed to new AMPA receptor clusters → late LTP
6. In parallel, repeated stimulation → local translation of PKMζ in dendrites → maintenance (§4.6)

- Early LTP differs between mossy fiber and Schaffer pathways; late LTP converges on the same cAMP–PKA–CREB route

Evidence: protein synthesis inhibitor spares early LTP but blocks late LTP; quantal analysis — early LTP raises release probability at one site, late LTP adds multiple release sites → late LTP is growth.

4.5 Synapse specificity: tagging and capture

Gene products are sent to all terminals, but only synapses **tagged** by recent activity can use them.

1. Strong stimulation (5 pulses of 5-HT) at synapse A → signal to the nucleus → CREB-dependent transcription

2. New mRNAs and proteins → fast axonal transport → delivered to all of the neuron's terminals
3. Weak stimulation (1 pulse) at synapse B → too little to turn on genes, but enough to mark B
4. Marked synapses use the delivered products → long-term facilitation and growth at A and B; unmarked synapses unchanged

- The mark has two parts:
  - Local PKA activity → needed to capture the products and start growth
  - PI3 kinase → triggers local translation of delivered mRNAs → needed to keep growth beyond the first day (without it, captured facilitation collapses)
- Same in hippocampus: one tetanus captures late LTP triggered elsewhere on the neuron

Evidence: one sensory neuron contacting two motor neurons — 5 pulses at one branch → long-term facilitation only there; adding 1 pulse at the other branch → long-term facilitation there too.

4.6 Maintenance: self-perpetuating molecules

Proteins turn over within days; memories last longer → maintenance needs a molecular state that sustains itself.

CPEB — a functional prion (Aplysia, flies)

1. Dormant mRNAs with short poly(A) tails are shipped to all terminals → not translated
2. 5-HT at a marked synapse (with PI3 kinase) → local synthesis of neuronal CPEB rises (no transcription needed)
3. CPEB passes a threshold concentration → its glutamine-rich, prion-like N-terminal domain switches it from a soluble, inactive monomer to an aggregated, active form
4. Aggregates recruit soluble CPEB and convert it → active state self-perpetuating

5. Active CPEB binds the CPE site of dormant mRNAs → recruits poly(A) polymerase → poly(A) tail lengthened
6. Polyadenylated mRNAs → recognized by ribosomes → translated locally (e.g., actin, tubulin, CPEB itself) → new synaptic structure stabilized at that synapse only

- Unlike pathogenic prions, the aggregated form does not kill cells

PKMζ (hippocampus)

1. Tetanus → enhanced local translation of PKMζ mRNA in CA1 dendrites
2. PKMζ = PKC isoform lacking the regulatory domain → needs no DAG or Ca2+ → constitutively active
3. Ongoing PKMζ activity → keeps extra AMPA receptors in the postsynaptic membrane → late LTP maintained

- **CaMKII:** a brief Ca2+ signal → autophosphorylation → Ca2+-independent activity → proposed switch

Evidence: a PKMζ inhibitor given after LTP or learning erases established late LTP and spatial memory → maintenance needs ongoing kinase activity.

5. Explicit memory: prefrontal cortex and hippocampus

The prefrontal cortex holds information briefly in **working memory** by persistent firing; the hippocampus stores explicit memories durably by synaptic change (LTP, §3–4), building a map of space.

5.1 Working memory — persistent activity in prefrontal cortex

- Paradigm: delayed match-to-sample → prefrontal neurons keep firing through the delay while the image is held in mind
- Contrast: long-term explicit storage relies on synaptic change, not ongoing firing

Intrinsic mechanism — a Ca2+/CAN channel positive-feedback loop

1. Acetylcholine → muscarinic receptors → signaling cascade that enables Ca2+-activated nonselective cation (CAN) channels

2. Brief stimulus → burst of action potentials → Ca2+ enters through voltage-gated Ca2+ channels
3. Ca2+ binds the cytoplasmic side of CAN channels → they open → inward cation current
4. Cytoplasmic Ca2+ stays high after the burst → inward current outlasts it → afterdepolarization
5. If the stimulus was strong enough → afterdepolarization reaches threshold → new spikes → more Ca2+ → more CAN current → firing sustains itself (rate graded with stimulus strength)

Network mechanisms

- Reverberation: recurrent excitatory connections keep each other firing, locally or across areas (prefrontal ↔ parietal ↔ inferotemporal)
- Mutual inhibition between two populations, each internally excitatory:
    1. Brief excitatory input to group 1 → group 1 fires more → more inhibition onto group 2
    2. Group 2 fires less → less inhibition onto group 1 (**disinhibition**)
    3. Group 1 fires even more → positive feedback → activity persists after the input ends
- Dopamine D1 receptors (→ Gs → cAMP): inverted-U — working memory best at intermediate activation

5.2 Spatial memory depends on hippocampal LTP

Manipulations that impair Schaffer LTP impair hippocampus-dependent spatial memory; those that enhance LTP enhance it.

- Paradigm: **Morris water maze** — hidden platform found via room cues (needs hippocampus); visible platform = control (does not)
- NMDA blocker or CA1-only NMDA knockout → no Schaffer LTP → poor spatial memory
- NMDA receptors needed to learn, not to recall → mirrors LTP induction vs maintenance (maintenance needs PKMζ, §4.6)

- Extra NR2B subunit → NMDA receptors pass more Ca2+ → larger LTP → faster learning
- Reversible transgenes (switched off in adults → LTP and memory restored) rule out a developmental artifact

Evidence: in vivo, learning strengthens Schaffer synapses and occludes later electrically induced LTP → learning itself induces LTP.

5.3 Cognitive map: place cells and grid cells

The hippocampus holds an **allocentric** (world-fixed) map: each place cell fires when the animal is in one location of a given environment, its place field.

1. Medial entorhinal **grid cells** → fire at many regularly spaced points forming a triangular grid → a landmark-independent coordinate system
2. Grid-cell output → perforant path → hippocampus
3. Hippocampal CA3/CA1 pyramidal cells transform it into **place fields**: each cell fires at one unique location
4. Population of place fields → map of that environment; forms within minutes in a new one

- Unlike sensory maps, not topographic or egocentric
- LTP needed for long-term **stability**, not formation: NMDA knockout → fuzzy fields; blocked late LTP (PKA inhibitor) → fields drift within hours

Stabilization needs attention

1. Animal attends to the space (e.g., foraging)
2. Dopamine → D1/D5 receptors → adenylyl cyclase → cAMP → PKA
3. PKA → late-LTP-type stabilization → place fields stable for days (without attention: stable only for hours)

- Needing attention makes the place-field map behave like explicit memory

5.4 Pattern separation vs pattern completion

- **Pattern separation** (tell similar episodes or contexts apart) → dentate gyrus (far more granule cells than CA3/CA1 cells); needs dentate NMDA receptors

and **adult-born** granule cells

- **Pattern completion** (recall a whole memory from partial cues) → recurrent CA3 ↔ CA3 connections; CA3 NMDA knockout impairs recall only when cues are reduced
- Adult neurogenesis is limited to dentate granule cells and olfactory bulb granule cells

6. Other implicit memory systems

The mechanisms of §2–4 recur across species and brain systems; only the transmitters, cells and circuits change.

6.1 Drosophila — olfactory fear conditioning

Same coincidence detection as Aplysia conditioning (§2.3), with **dopamine** in place of serotonin as the US signal.

- Paradigm: odor (CS) paired with shock (US) → flies avoid that odor in a choice maze

1. Odor (CS) → antennal lobes → Kenyon cells of the mushroom bodies → Ca2+ rises in Kenyon cells
2. Shock (US) → dopaminergic neurons fire → dopamine released onto the Kenyon cells
3. Dopamine → metabotropic receptor → Gs → stimulates a Ca2+/calmodulin-dependent adenylyl cyclase (*rutabaga*)
4. Cyclase gets both signals — Ca2+ (CS) and Gs (US) → synergistic, large cAMP rise → PKA → short-term memory
5. With repetition → CREB-dependent gene expression → long-term memory

- Learning mutants all hit the cAMP pathway: *dunce* (phosphodiesterase that degrades cAMP), *rutabaga* (the cyclase), *amnesiac* (peptide that stimulates the cyclase)
- CREB needed selectively for long-term memory; extra CREB → long-term memory from training that normally gives only short-term

- CPEB prion-like domain needed for long-term memory (§4.6)

Evidence: activating dopamine neurons directly (instead of shock) while presenting an odor produces avoidance → dopamine carries the aversive US.

6.2 Amygdala — learned fear

CS and US signals converge in the **lateral amygdala**, where pairing produces LTP (as in §3.2) with a CREB-dependent long-term phase (§4).

1. Tone (CS) and shock (US) → auditory and somatosensory thalamus → lateral nucleus, both by a fast **direct** path and a slower **indirect** path through cortex
2. Converging strong input → large postsynaptic depolarization → Ca2+ enters through NMDA receptors and L-type voltage-gated Ca2+ channels
3. Ca2+ → kinase cascade → AMPA receptors inserted postsynaptically + more transmitter release presynaptically → tone input potentiated (LTP)
4. Persistence: PKA + MAPK → CREB → gene expression (§4)
5. Potentiated lateral nucleus → central nucleus (output) → hypothalamus and brainstem → tone alone now evokes fear (freezing, heart rate, other autonomic changes)

- Neurons with more CREB at the time of learning are preferentially recruited into the memory

Evidence: fear conditioning occludes later LTP in amygdala slices, and blocking AMPA-receptor insertion in a minority of lateral-nucleus neurons weakens fear memory → learned fear uses LTP.

6.3 Striatum — habits

Habits are stimulus-response associations acquired gradually by repetition and run without conscious control; they depend on the striatum (caudate).

- Complex habits are built up from simpler stimulus-response links
- Likely storage site: cortex → striatum synapses, which show NMDA-dependent LTP; its persistence and habit memory need CREB

Evidence: eight-arm maze double dissociation — *win-shift* (remember which arms you visited today = episodic) needs hippocampus, not caudate; *win-stay* (lit arms

always hold food = regularity) needs caudate, not hippocampus.

7. Cerebellum: motor control and motor learning

The cerebellum does not initiate movement; it makes movement accurate. It compares intended with actual movement, sends feed-forward corrections to the descending motor systems, and uses internal models of the body that it tunes by learning. One uniform microcircuit performs the same computation on different inputs in each region.

7.1 What lesions reveal

Cerebellar damage **disrupts** movement rather than abolishing it (contrast: cortical damage → paralysis).

- **Hypotonia:** less resistance to passive movement; "pendular" reflexes that swing several times
- **Astasia-abasia:** cannot hold posture or stand upright → wide-based stance, falls
- **Ataxia** — incoordination of multijoint voluntary movement:
  - Delayed initiation
  - **Dysmetria:** wrong range and direction
  - **Dysdiadochokinesia:** irregular rapid alternating movements
  - **Decomposition of movement:** joints moved one at a time; proximal joints not braced
- **Action (intention) tremor** at the end of movement: mistimed antagonist braking (§7.4)
- Loss of automaticity: each movement must be "thought out" → cortex takes over the details
- Deficits are **ipsilateral** (outputs cross twice, §7.2)

7.2 Three functional regions

Same circuit, different inputs and output targets.

Vestibulocerebellum (flocculonodular lobe; oldest) — balance and eye movements

1. Vestibular nuclei (head motion, position relative to gravity) + visual input → mossy fibers → flocculonodular lobe
2. Purkinje cells → project **directly** to the vestibular nuclei, bypassing the deep nuclei
3. Midline Purkinje cells → lateral vestibular nucleus → vestibulospinal tracts → axial muscles and limb extensors → balance in stance and gait
4. Lateral Purkinje cells → medial vestibular nucleus → eye movements: VOR, smooth pursuit, head–eye coordination

- Acts only when the brainstem reflex alone fails: Purkinje output changes only when eye motion differs from what the VOR predicts
- Lesion → poor pursuit toward the lesioned side; VOR can't be suppressed by fixation

Spinocerebellum, vermis — posture, locomotion, eye movements

1. Somatosensory input from head and proximal body (+ visual, auditory, vestibular) → vermis
2. Vermis Purkinje cells → fastigial nucleus
3. Fastigial nucleus → bilaterally to reticular formation and lateral vestibular nuclei
4. These → reticulospinal and vestibulospinal tracts (medial descending systems) → neck, trunk, proximal limb muscles

- Vermis also controls saccades and smooth pursuit (via a small part of the fastigial nucleus → brainstem saccade generator)
- Modulates the descending commands rather than issuing them

Spinocerebellum, intermediate hemispheres — distal limbs

1. Somatosensory input from the limbs (spinocerebellar tracts) → intermediate hemispheres
2. Purkinje cells → interposed nucleus

3. Interposed axons → superior cerebellar peduncle → cross the midline → magnocellular red nucleus
4. Red nucleus axons → cross back → rubrospinal tract → spinal cord on the original side → distal limb muscles
5. Other interposed axons → contralateral ventrolateral thalamus → limb areas of motor cortex → corticospinal tract (crosses at pyramidal decussation) → original side

- Two midline crossings → a cerebellar lesion disrupts the **ipsilateral** limbs

Cerebrocerebellum (lateral hemispheres; newest, enlarged in humans) — planning

1. Cerebral cortex → pontine nuclei → middle cerebellar peduncle → contralateral lateral hemisphere (its only input)
2. Purkinje cells → dentate nucleus
3. Dentate → superior peduncle → contralateral ventrolateral thalamus → premotor and motor cortex (also prefrontal)
4. Dentate → parvocellular red nucleus → inferior olive → climbing fibers back to the cerebellum (a loop)

7.3 The microcircuit

Two excitatory inputs converge on **Purkinje cells**, the sole output of the cortex, which inhibit the deep nuclei; the deep nuclei weigh that inhibition against direct excitation.

- Layers:
  - Granular (input): very numerous granule cells, Golgi cells; mossy fibers end here in glomeruli
  - Purkinje cell layer (output): Purkinje cell bodies; GABAergic axons → deep nuclei / vestibular nuclei
  - Molecular (processing): flat, fan-like Purkinje dendrites, parallel fibers, stellate and basket cells

Mossy fiber → parallel fiber pathway (rate code)

1. Mossy fibers (from spinal cord, brainstem, and cortex via pons) → excite granule cells in glomeruli
2. Granule-cell axons ascend → split into parallel fibers running along the folium
3. Parallel fibers run perpendicular to the flat Purkinje dendrites → each fiber contacts many Purkinje cells, each weakly → tiny EPSPs
4. Many parallel fibers summed → **simple spikes**; their rate follows mossy input → encodes magnitude and duration of stimuli and movements

Climbing fiber pathway (event signal)

1. Inferior olive neuron fires → its climbing fiber, wrapped around one Purkinje cell's soma and proximal dendrites, releases glutamate at many synapses
2. Strong depolarization → opens voltage-gated Ca2+ channels in soma and dendrites → prolonged depolarization
3. Prolonged depolarization → **complex spike**: large action potential + high-frequency burst of smaller spikes
4. Climbing fibers fire rarely and in relation to specific events → signal *that* something happened, not how much; olive cells are electrically coupled → synchronous firing

- One climbing fiber per Purkinje cell, arranged in parasagittal strips → precise control of functionally related Purkinje cells

Output: comparing excitation and inhibition

1. Mossy and climbing fiber collaterals → directly excite deep-nuclear neurons
2. Same signals, processed by the cortex → Purkinje cells → GABA → inhibit those deep-nuclear neurons
3. Net deep-nuclear output = excitation sculpted by Purkinje inhibition

- Within the cortex, parallel fibers excite Purkinje cells but also inhibit them via interneurons: stellate cells (local, same parallel-fiber beam) and basket cells (flanking beams → lateral inhibition onto the cell body)

Recurrent loops

- Cortex → pons → cerebellum → thalamus → the same cortical area (parallel closed loops)
- Olive loop:
    1. GABAergic deep-nuclear neurons → inhibit the inferior olive
    2. More nuclear inhibition → olive fires less → less climbing-fiber input to that cerebellar region
    3. → each region regulates its own climbing-fiber (teaching) input; inhibition can also decouple olive cells
- Golgi loop:
    1. Parallel fibers → excite Golgi cells
    2. Golgi cells → GABA onto granule cells
    3. Mossy excitation of granule cells suppressed → only few granule cells active at a time (sparse code)

7.4 Control principles

Comparing intended with actual movement

1. Dorsal spinocerebellar tract → reports actual movement from muscle and joint receptors (passive or active)
2. Ventral spinocerebellar tract → receives the same inputs as spinal motor neurons → reports an **efference copy** of the motor command (active movement only)
3. Cerebellum compares the two → determines whether the command must be modified to achieve the intended movement

- Cerebellar neurons fire with direction and speed, somatotopically, in step with motor cortex → works in loops with cortex

Feed-forward timing of agonist and antagonist

1. Agonist contracts → limb accelerates toward the target
2. Cerebellar output (interposed, dentate) → triggers the antagonist burst early, before sensory feedback could arrive

3. Antagonist brakes → limb stops on target

When the deep nuclei are inactivated:

1. Agonist activation becomes slower and prolonged
2. Antagonist is delayed until the limb has overshot
3. Sensory feedback drives a correction → the correction is itself mistimed → new error
4. Repeated delayed corrections → oscillation around the target = intention tremor

Internal models

- Kinematic and dynamic models of limbs and eyes turn a desired endpoint into properly timed and scaled commands
- Interaction torques:
    1. Moving the forearm → its momentum creates forces that move the upper arm
    2. Internal model predicts these → shoulder muscles contract in advance to cancel them
    3. Cerebellar damage → prediction fails → shoulder perturbed → multijoint movements less accurate than single-joint
- Regularity matters: excessively variable Purkinje firing (even at normal mean rate) → ataxia

Evidence: decerebrate cat walking with dorsal roots cut — dorsal tract falls silent, ventral tract stays rhythmic → ventral tract carries the internal command.

7.5 Cerebrocerebellum: planning, timing, cognition

- Dentate neurons fire before movement, before motor cortex → planning; inactivation delays but does not prevent movement
- Lesions → variable delays in starting movements, irregular timing of components

- Lateral lesions disrupt the **timing** of serial events, even purely perceptual ones (judging durations or speeds); medial lesions disrupt execution
- Active in cognitive tasks (word association, puzzle solving); dentate region linked to cortical working-memory area
- Premotor → lateral cerebellum → red nucleus → olive loop may support mental rehearsal of movement

7.6 Motor learning and cerebellar LTD

Marr–Albus–Ito theory: climbing fibers carry an **error signal**; when they fire together with parallel fibers, those parallel fiber → Purkinje synapses undergo **long-term depression (LTD)**, suppressing the inputs that produced the error.

The learning logic

1. Movement command and context → mossy fibers → granule cells → parallel fibers → Purkinje cells → deep nuclei → movement
2. Movement errs → climbing fiber fires (error signal, e.g., retinal slip in the VOR)
3. Parallel-fiber synapses active at that moment → depressed (LTD)
4. Next attempt → those inputs drive the Purkinje cell less → altered simple-spike pattern → altered deep-nuclear output → movement corrected
5. Error disappears → climbing fiber stops signaling → learning stops

LTD induction — coincidence of two inputs at the Purkinje cell

1. Parallel fibers → glutamate → AMPA receptors and metabotropic mGluR1 on the Purkinje spine
2. mGluR1 → Gq → phospholipase C → IP3 + DAG
3. IP3 → IP3 receptors on the endoplasmic reticulum → Ca2+ released from internal stores
4. Climbing fiber → strong depolarization → Ca2+ influx through voltage-gated Ca2+ channels
5. Ca2+ (from both sources) + DAG → activate PKC — the convergence point of the two inputs

6. PKC phosphorylates the GluA2 subunit of AMPA receptors → receptor releases its anchoring protein (GRIP) and binds PICK1
7. → clathrin-mediated endocytosis of AMPA receptors → fewer receptors → reduced response to glutamate = LTD

- Climbing-fiber Ca2+ alone is normally not enough: mGluR1 signaling must coincide
- Contrast with hippocampal LTD (§3.4): there a *small* Ca2+ signal → phosphatases; here coincident climbing + parallel fiber input → kinase (PKC). Both end in AMPA endocytosis
- A late phase needs transcription and protein synthesis (as in §4)

Model paradigms

- **VOR adaptation:** magnifying/minifying glasses → reflex gain adjusts; climbing fibers signal retinal slip; needs flocculus
- **Eyeblink conditioning:** tone (CS) + air puff (US) → correctly timed blink to tone alone
- **Prism adaptation:** shifted dart throws corrected with practice, with an aftereffect when prisms are removed; fails with cerebellar cortex or olive damage

Beyond LTD

- Learning occurs at **multiple sites**: deep nuclei (mossy fiber → nuclear synapses, shaped by Purkinje input), parallel-fiber LTP (needed for bidirectional change), potentiation of inhibitory synapses onto Purkinje cells
- LTD's necessity is debated: some LTD-deficient mice learn normally → other plasticity may compensate
- Overall role: continuously tune internal models so commands become fast, accurate, smooth

Evidence: in vitro, parallel fibers stimulated together with climbing fibers are depressed, unpaired ones are not → associative, input-specific. In monkeys adapting to a new load, climbing fibers fire on every trial during adaptation, then fall silent as simple spikes decrease → the predicted sequence.

8. Experience reshapes the brain; common building blocks

Learning-induced synaptic change scales up to cortical maps, so each brain is uniquely modified by its experience — together with its genes, the biological basis of **individuality**.

- Use expands representation: monkeys repeatedly using a few fingertips → enlarged somatosensory representation of those fingertips
- Correlated input shapes borders: two fingers surgically joined → the boundary between their cortical zones disappears (LTP-like associative tuning)
- String players: larger representation of the fingering (left) hand, most when training began in childhood

Common principles across all systems:

- Associative plasticity is built from nonassociative components in the same cell (sensitization → conditioning)
- Coincidence is detected by single proteins: adenylyl cyclase (Ca^{2+} + modulator) and the NMDA receptor (glutamate + depolarization)
- Associative steps act in series: postsynaptic detection + retrograde signal to active terminals only
- Short-term = modification of existing proteins; long-term = PKA → CREB → chromatin → gene expression → structural change, captured by tagged synapses and kept by self-perpetuating molecules